

Being Mosquitone: Illusory Deep Intrusion of Mosquito into the Head driven by HMD-Based Audiovisual Integration

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Figure 1: Being Mosquitone

Abstract

"Being Mosquitone" is a head-mounted display (HMD) system that creates a compelling illusion of a mosquito directly entering the interior of the head through the ears. This bizarre sensation arises from the combination of a mosquito sound signal, played through either a mobile speaker or headphones, and the visual representation of the mosquito flying around or rotating inside the participant's semi-transparent head avatar, synchronized with their own physical head tilt, as presented via the HMD. The key aspect of the experience is the synchronization between the mosquito's visual position inside the head and the left-right panning of the stereo audio played through the headphones. The system was tested in a public exhibition with 123 participants, followed by a post-experience questionnaire. The results showed that more than 70% of participants reported experiencing the illusory sensation perceived inside the head as "strong" or "very strong," and over 85% provided positive feedback. Additionally, a controlled experiment revealed that the

illusion's effect dramatically decreased when only visual stimuli were presented, and that the synchronization of audiovisual stimuli (left-right congruence) significantly enhanced the illusion.

CCS Concepts

• **Human-centered computing** → **Virtual reality**.

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1 Background

The perception of sound through headphones or earphones involves two primary phenomena: intra-head localization and external localization. Intra-head localization occurs when sound is perceived as originating from within the head, often near the center or around the ears, which commonly happens during stereo audio playback. This effect arises because sound positioning relies solely on interaural time differences (ITD) and interaural level differences (ILD), without additional spatial cues. In contrast, external localization occurs when sound is perceived as originating from outside the

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head, typically achieved through binaural recordings or the application of head-related transfer functions (HRTF), which provide more complex spatial cues based on the listener's anatomy[4][1].

In most acoustic research and audio design, intra-head localization is considered undesirable due to its unnatural spatial characteristics, as it fails to replicate the real-world perception of sound distance and directionality. However, intra-head localization can also be reframed as a powerful perceptual tool when applied intentionally. In the context of virtual reality (VR), it may be particularly useful in enhancing the illusory sensation of something being present inside the head, contributing to unique sensory experiences. In fact, our research team previously introduced XRAYHEAD[2], a system that creates the illusory sensation of the skull inside the scalp being directly touched through the integration of visual and tactile stimuli. By appropriately integrating auditory signals, the control of such illusions may become more feasible.

This study explores how intra-head localization can be intentionally leveraged to create a compelling sensation of intrusion into the head using a VR system, emphasizing the potential of this auditory phenomenon for novel immersive applications.

2 Interaction Design

Being Mosquitone consists of an HMD (HTC Vive) with integrated headphones and a mobile speaker (MIATONE QBOX Bluetooth Speaker) with a Vive Tracker simply attached for position tracking. Participants sit in a chair and wear the HMD, through which the display reveals a transparent view of the interior, showing a CG model of the head with the ears connected (head avatar). This head avatar moves in response to the participant's head tilt, thereby achieving a basic full-body illusion. [5][3]

2.1 Extra-head mode

Initially, the mosquito in the HMD flies around the outside of the head. The position of the mosquito in the HMD corresponds to the location of the mobile speaker, which is held and moved by the experimenter. The sound of the mosquito, obtained from royalty-free sources, is played through the mobile speaker, simulating the real-world experience of a mosquito flying around the participant's head. During this phase, no sound is played through the headphones.

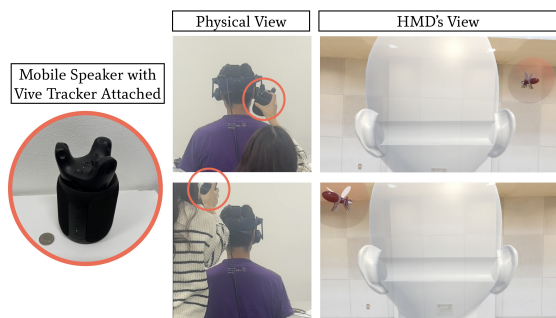


Figure 2: View of Extra-head mode

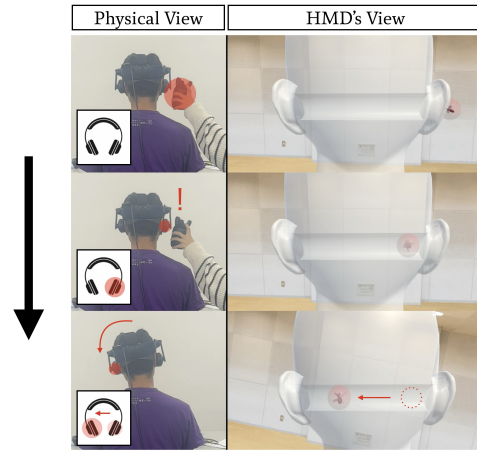


Figure 3: Switch from Extra-Head to Intra-Head Mode

2.2 Intra-head mode

Subsequently, the experimenter brings the mobile speaker as close as possible to either the participant's left or right ear. The mosquito sound output is then switched from the mobile speaker to the headphones worn by the participant. At this moment, within the HMD environment, the mosquito, previously positioned near the head-avatar's ear, becomes encapsulated in a sphere and instantaneously shifts to either the left or right end of a tunnel connecting both ears inside the head. When the participant tilts their head laterally, the capsule containing the mosquito rolls in the corresponding direction, with the ends of the tunnel acting as barriers (visuo-proprioception synchronization). Notably, the mosquito's position within the tunnel is synchronized with the left-right balance of the headphone audio output, such that the perceived volume increases on the side toward which the participant tilts their head (visuo-auditory synchronization).

Finally, after the participant maintains a head tilt in one direction for a certain period, the capsule exits the ear and the mosquito falls freely to the ground. At this moment, the headphone output ceases, resulting in complete silence.

3 Participant Responses in the Open Exhibition

The proposed system was first presented to the public during a laboratory exhibition in November 2024. Notably, many participants reported a pronounced sense of discomfort at the moment the mosquito appeared to enter their ear.

During the exhibition, a questionnaire-based experiment was conducted following a few minutes of experience with the system. Participants responded to four questions: (Q1) "I felt as if a mosquito was entering the interior of my head," (Q2) "I felt as if both ears were being penetrated," (Q3) "I felt as if the mosquito was moving between my ears," and (Q4) "I felt as if the mosquito sound was occurring inside my head."

Responses were collected using a 7-point Likert scale ranging from -3 to +3. The proportions of participants who rated each item as +2 ('strong') or +3 ('very strong') were 72%, 49%, 75%, and 71%,

respectively (Figure 4). Furthermore, 87% of participants gave positive ratings (a score of +1 or higher) for Q1, Q2, and Q4 regarding the illusion. These results demonstrate that the proposed system effectively produces a strong illusory sensation within the head.

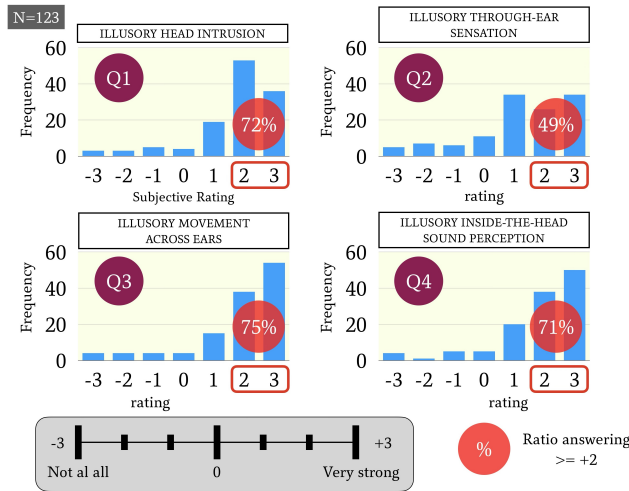


Figure 4: Experimental Results of the Public Exhibition Experiment

4 Laboratory Experiment

4.1 Procedure

In an additional experiment, a study involving 12 participants was conducted to examine the effects of audiovisual synchronization and gravity direction on the sensation of illusory inside-head sensation (Q1 to Q3). Specifically, as a factor of audiovisual synchronization, a visual-only condition (MUTE) and a condition in which the left-right audio balance was reversed relative to the mosquito's position (PAN ASYNC) were introduced in addition to the standard condition (PAN SYNC). As a factor of gravity direction, a condition where gravity was reversed (GRAVITY UPWARD) was given in addition to the standard gravity condition (GRAVITY DOWNWARD). Note that, under the GRAVITY UPWARD condition, the capsule rolls upward in the opposite direction of the head tilt when the head is tilted to the left or right.

For these six conditions (2x3), participants sequentially experienced a transition from the extra-head mode to the intra-head mode. They then performed a task of rolling the mosquito capsule left and right for approximately 30 seconds, followed by answering four questionnaire items.

4.2 Results

A two-factor analysis of variance (ANOVA) with aligned rank transformation was conducted. No significant main effects were observed for the gravity direction factor across any of the questionnaire items, while a significant main effect was found for the audiovisual synchronization's factor. Multiple comparisons revealed a significant

increase in the illusion effect due to audiovisual panning synchronization in Q2 (illusory through-ear sensation) and Q3 (illusory movement across ears).

The results showed that the presence of sound dramatically enhanced the intensity of the inside-head illusion. While the effect of audio panning synchronization was significant, as expected, the absence of a detectable effect from the gravity direction factor was entirely unexpected. In the GRAVITY UPWARD condition, some participants reported the impression of water rising inside their ears. This suggests that in the gravity-inversion mode, participants do not necessarily alter their perception of gravity itself. Rather, they rely on consistent directional feedback from head tilting as a cue and reinterpret the attributes of various CG objects while maintaining physical laws.

References

- [1] Kazuhiro Iida. 2019. *Head-related transfer function and acoustic virtual reality*. doi:10.1007/978-981-13-9745-5
- [2] Kento Imai, Haruka Kayano, and Kenri Kodaka. 2022. XRAYHEAD. *Proceedings - SIGGRAPH Asia 2022 XR, SA 2022* (11 2022). doi:10.1145/3550472.3558411
- [3] Vilayanur S. Ramachandran, Beatrix Krause, and Laura K. Case. 2011. The phantom head. *Perception* 40 (2011). Issue 3. doi:10.1068/p6754
- [4] M. Risoud, J. N. Hanson, F. Gauvrit, C. Renard, P. E. Lemesre, N. X. Bonne, and C. Vincent. 2018. Sound source localization. Issue 4. doi:10.1016/j.anorl.2018.04.009
- [5] Roy Salomon, Melanie Lim, Christian Pfeiffer, Roger Gassert, and Olaf Blanke. 2013. Full body illusion is associated with widespread skin temperature reduction. *Frontiers in behavioral neuroscience* 7 (jan 2013), 65. doi:10.3389/fnbeh.2013.00065

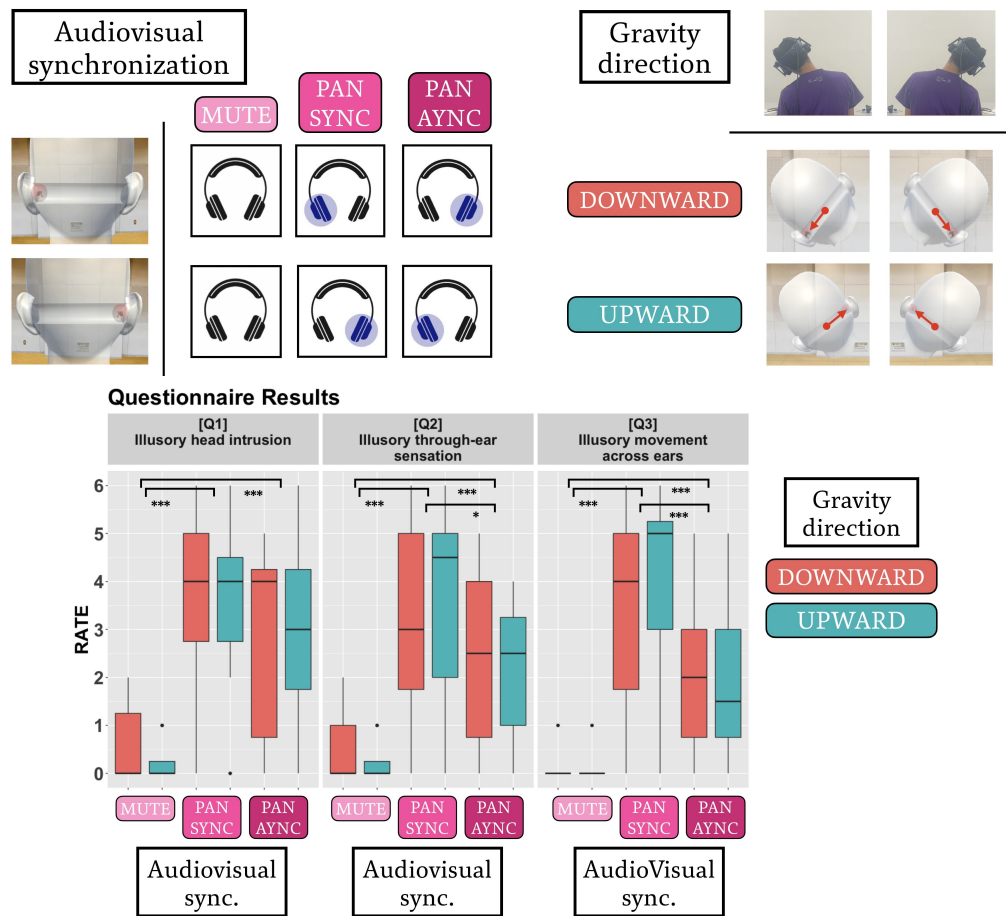


Figure 5: TOP: Experimental Factors of the Laboratory Experiment. BOTTOM: Experimental Results of the Laboratory Experiment. Statistical symbols indicate significant differences based on multiple comparisons of visuo-audio conditions (*: $p < 0.001$, *: $p < 0.05$)**